SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
ProgramName: B. Tech		**Assignment Type: Lab**	**AcademicYear:** 2025-2026
CourseCoordinatorName		Venkataramana Veeramsetty	
Instructor(s)Name		Dr. V. Venkataramana (Co-ordinator) Dr. T. Sampath Kumar Dr. Pramoda Patro Dr. Brij Kishor Tiwari Dr.J.Ravichander Dr. Mohammand Ali Shaik Dr. Anirodh Kumar Mr. S.Naresh Kumar Dr. RAJESH VELPULA Mr. Kundhan Kumar Ms. Ch.Rajitha Mr. M Prakash Mr. B.Raju Intern 1 (Dharma teja) Intern 2 (Sai Prasad) Intern 3 (Sowmya) NS_2 (Mounika)	
CourseCode	24CS002PC215	**CourseTitle**	AI Assisted Coding
Year/Sem	II/I	**Regulation**	R24
Date and Day of Assignment	Week3 - Wednesday	**Time(s)**	
Duration	2 Hours	**Applicableto Batches**	
AssignmentNumber: **6.3**(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*ExpectedTime to complete*
1	Lab 6: AI-Based Code Completion – Classes, Loops, and Conditionals **Lab Objectives:** • To explore AI-powered auto-completion features for core Python constructs. • To analyze how AI suggests logic for class definitions, loops, and conditionals. • To evaluate the completeness and correctness of code generated by AI assistants. **Lab Outcomes (LOs):**	Week3 - Wednesday

After completing this lab, students will be able to:

- Use AI tools to generate and complete class definitions and methods.
- Understand and assess AI-suggested loops for iterative tasks.
- Generate conditional statements through prompt-driven suggestions.
- Critically evaluate AI-assisted code for correctness and clarity.

Task Description#1 (Classes)

- Use AI to complete a Student class with attributes and a method.
- Check output
- Analyze the code generated by AI tool

Instructions:

- **Initialize class with attributes like name, roll no, marks**
- **Method to display student details**
- **Method to calculate grade based on marks (A:>=90, B: >=75, C: >=60, else Fail)**

Start Writing code and auto complete using any AI tool

Expected Output#1

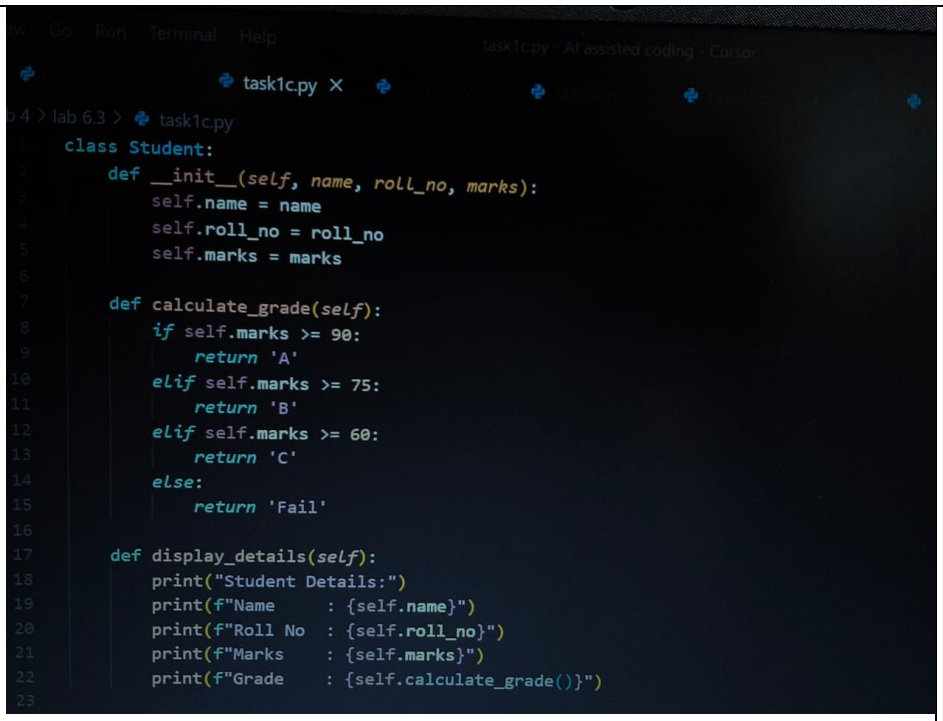
- Class with constructor and display_details() method

PROMPT :

```
class Student:
    def __init__(self, name, roll_no, marks):
        self.name = name
        self.roll_no = roll_no
        self.marks = marks

    def display_details(self):
        print("Student Details:")
        print(f"Name     : {self.name}")
        print(f"Roll No  : {self.roll_no}")
        print(f"Marks    : {self.marks}")
        print(f"Grade    : {self.calculate_grade()}")
```

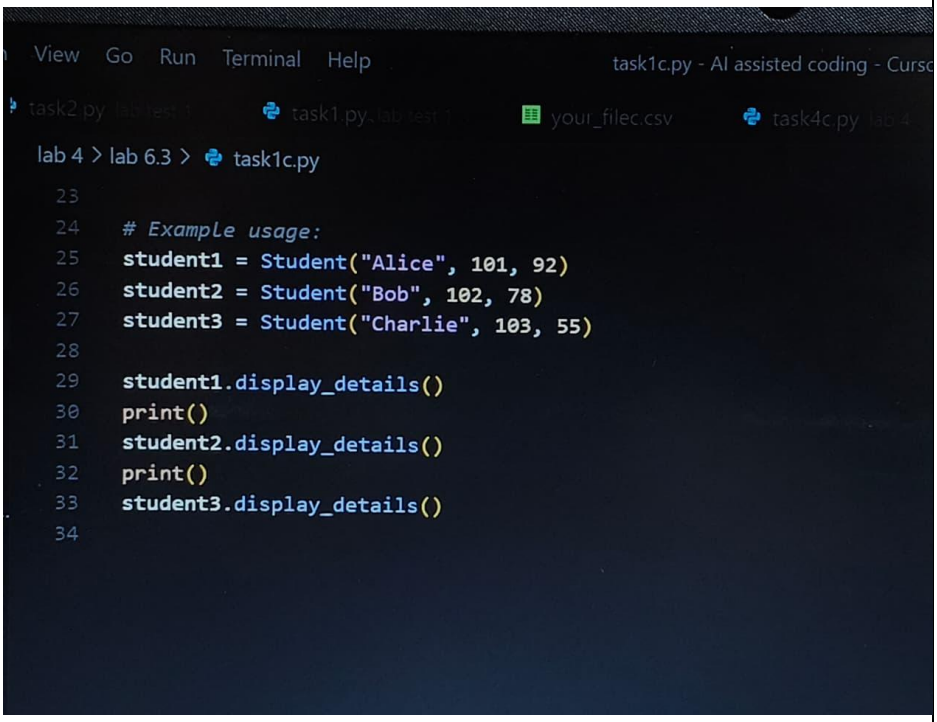
Now complete the remaining code and Initialize class with attributes like name, roll no, marks, Method to display student details, Method to calculate grade based on marks (A:>=90, B: >=75, C: >=60, else Fail) and display the output.



```python
class Student:
    def __init__(self, name, roll_no, marks):
        self.name = name
        self.roll_no = roll_no
        self.marks = marks

    def calculate_grade(self):
        if self.marks >= 90:
            return 'A'
        elif self.marks >= 75:
            return 'B'
        elif self.marks >= 60:
            return 'C'
        else:
            return 'Fail'

    def display_details(self):
        print("Student Details:")
        print(f"Name      : {self.name}")
        print(f"Roll No   : {self.roll_no}")
        print(f"Marks     : {self.marks}")
        print(f"Grade     : {self.calculate_grade()}")
```



```python
# Example usage:
student1 = Student("Alice", 101, 92)
student2 = Student("Bob", 102, 78)
student3 = Student("Charlie", 103, 55)

student1.display_details()
print()
student2.display_details()
print()
student3.display_details()
```

```
print()
```

```
Problems  Output  Debug Console  Terminal  Ports

... hon313\python.exe' 'c:\Users\kanch\.cursor\extensions\ms-python.debugpy
bugpy\launcher' '49890' '--' 'c:\Users\kanch\OneDrive\Desktop\AI assist
Student Details:
Name      : Alice
Roll No   : 101
Marks     : 92
Grade     : A

Student Details:
Name      : Bob
Roll No   : 102
Marks     : 78
Grade     : B

Student Details:
Name      : Charlie
Roll No   : 103
Marks     : 55
Grade     : Fail
PS C:\Users\kanch\OneDrive\Desktop\AI assisted coding>
                                        Ctrl+K to generate a command
```

Task Description#2 (Loops)

- Prompt AI to complete a function that prints the first 10 multiples of a number using a loop.
- Analyze the generated code
- Ask AI to generate code using other controlled looping

Write code using **For** Loop, later complete code using **While** Loop

Expected Output#2

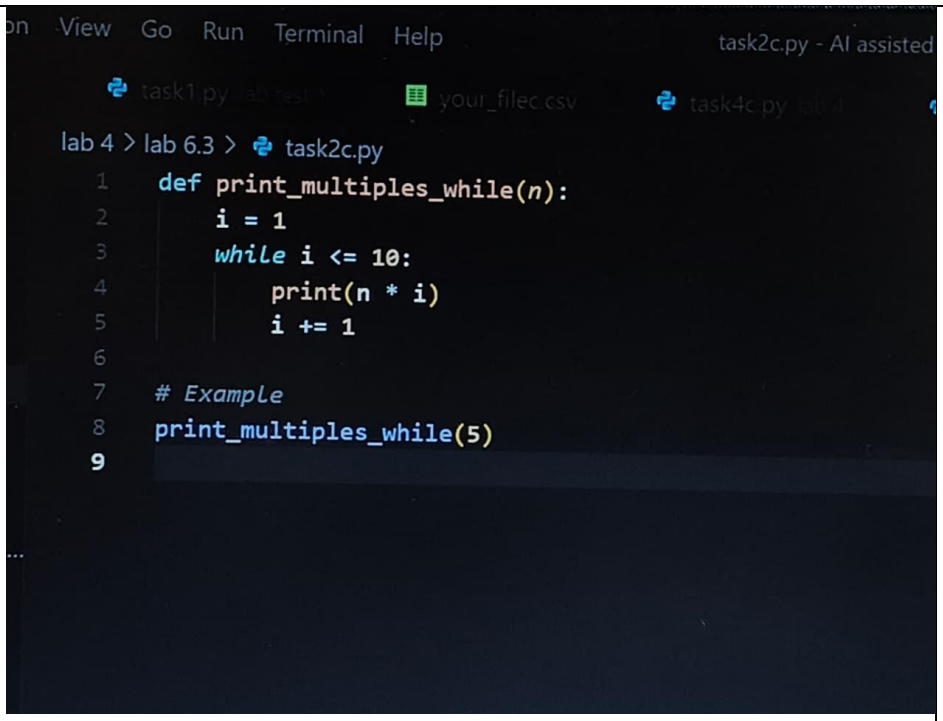
- Correct loop-based implementation

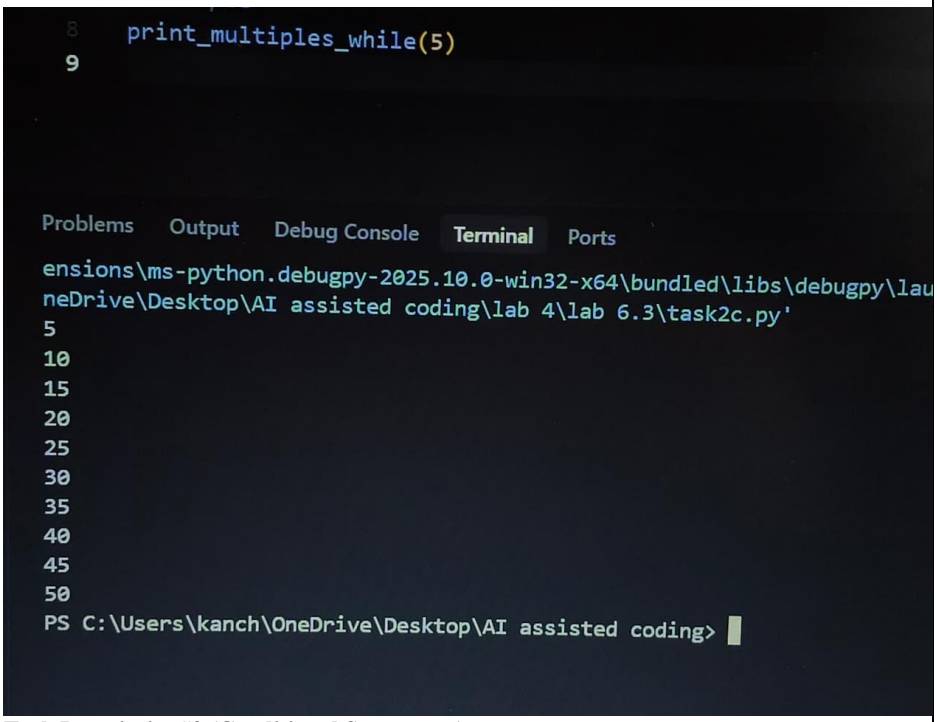
PROMPT:

```
def print_multiples(n):
    for i in range(1, 11):
        print(n * i)

# Example
print_multiples(5)
```

Now generate the above code using while loop.





Task Description#3 (Conditional Statements)

- Ask AI to write nested if-elif-else conditionals to classify age groups.
- Analyze the generated code
- Ask AI to generate code using other conditional statements

Table: Age Group Classification Logic

Age Range	Age Group
0 – 12 years	Child
13 – 19 years	Teen
20 – 59 years	Adult
60 years & above	Senior

-

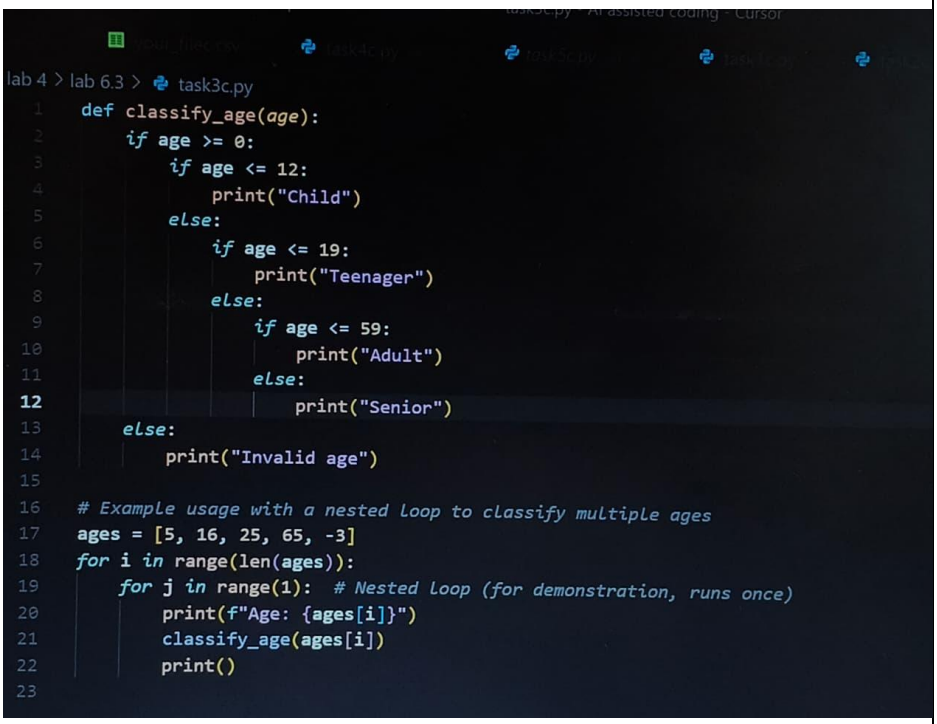
Expected Output#3

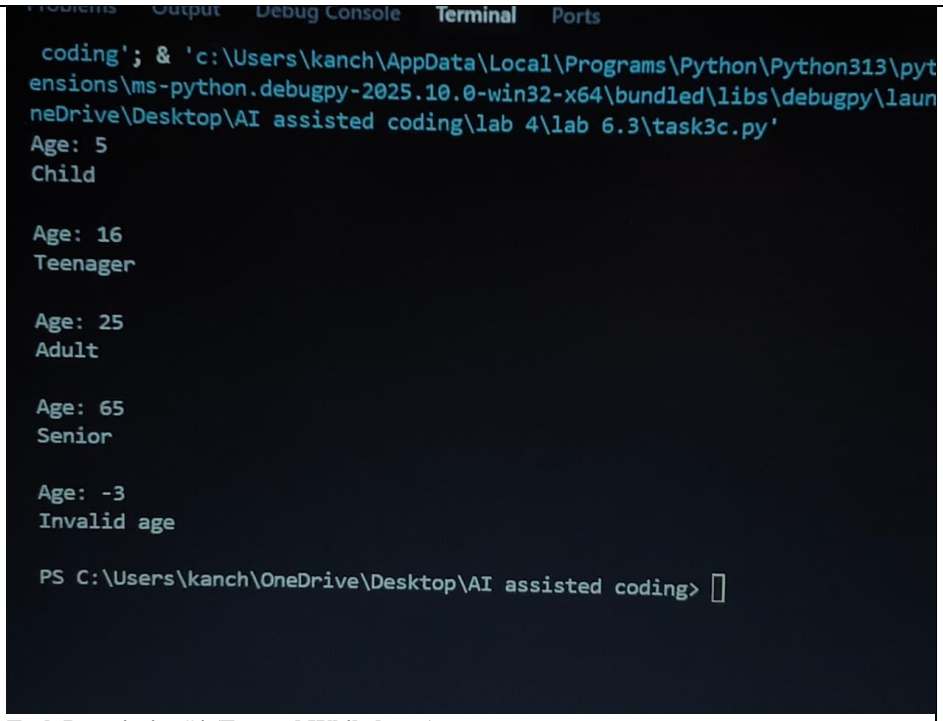
- Age classification function with appropriate conditions and with explanation

PROMPT:

def classify_age(age):
if age >= 0:
if age <= 12:
print("Child")
elif age <= 19:
print("Teenager")
elif age <= 59:
print("Adult")

Complete the full code using nested loop.





Task Description#4 (For and While loops)

- Generate a sum_to_n() function to calculate sum of first n numbers
- Analyze the generated code
- Get suggestions from AI with other controlled looping

Expected Output#4

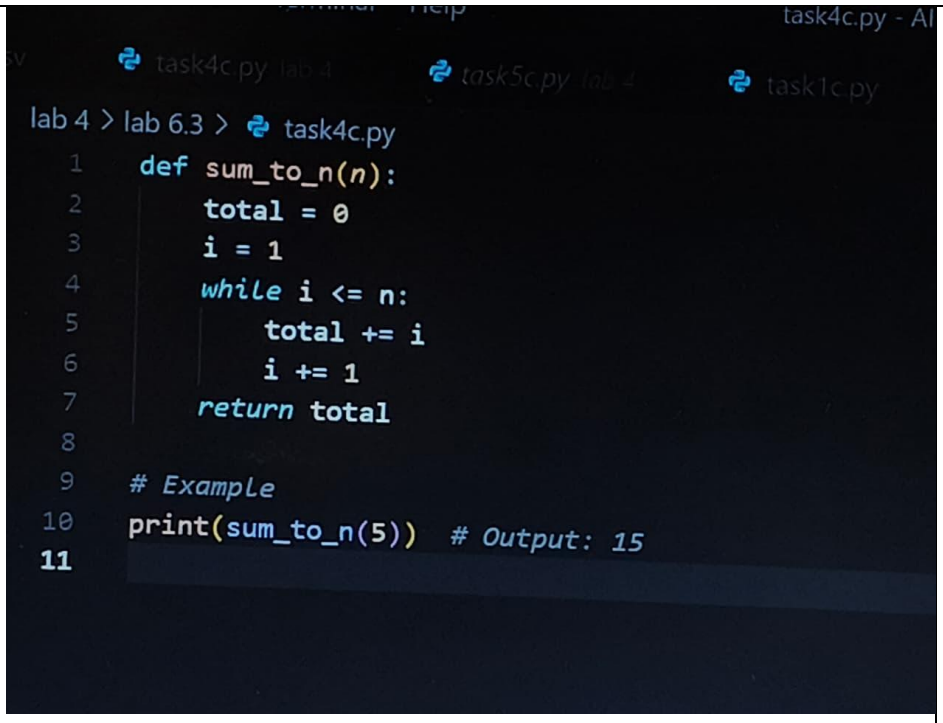
- Python code with explanation

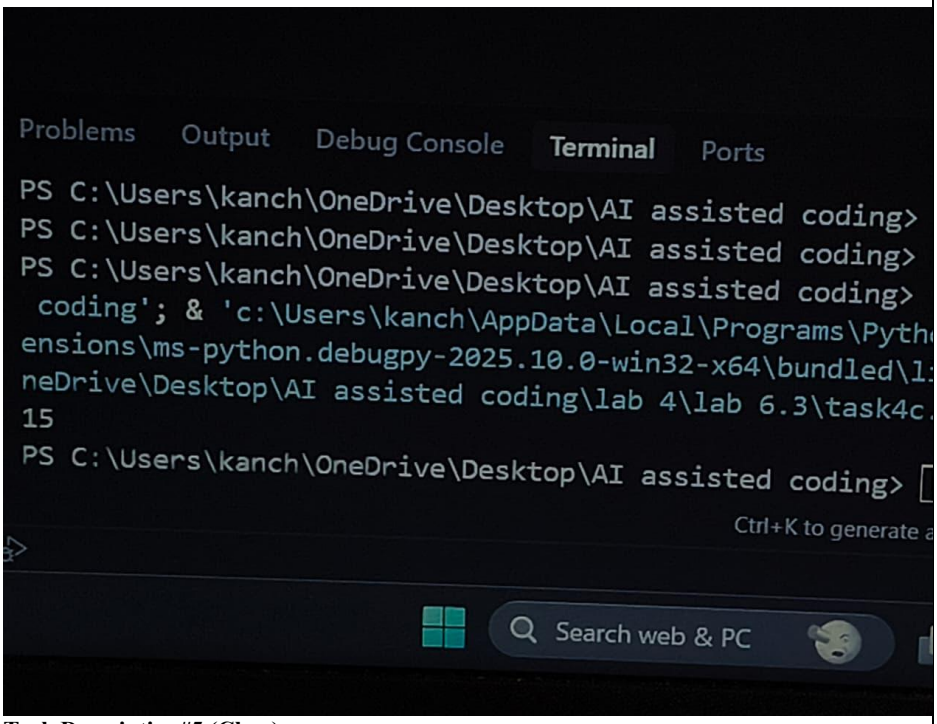
PROMPT:

```
def sum_to_n(n):
    total = 0
    for i in range(1, n + 1):
        total += i
    return total

# Example
print(sum_to_n(5))  # Output: 15
```

Generate the above code using while loop.





Task Description#5 (Class)

- Use AI to build a BankAccount class with deposit, withdraw, and balance methods.
- Analyze the generated code
- Add comments and explain code

Instructions

- **Initialize BankAccount class with attributes like name, balance**
- **Method to deposit amount**
- **Method to withdraw amount**

- **Method to check balance**

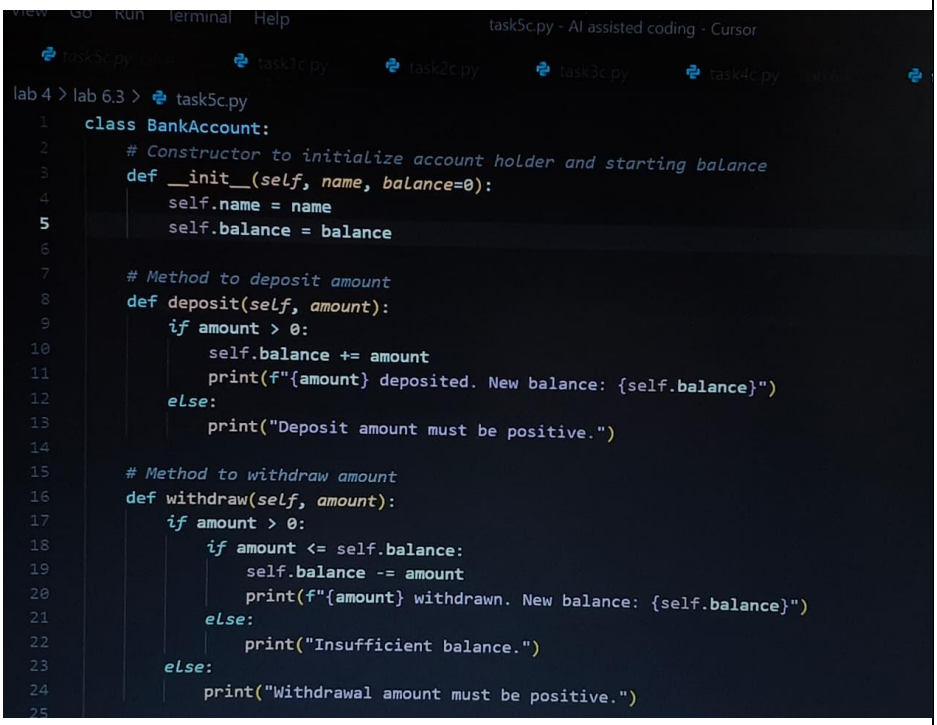
Expected Output#5

- Python code with explanation

PROMPT :

class BankAccount:
 # *Constructor to initialize account holder and starting balance*
def __init__(*self*, *name*, *balance*=0):
self.name = name
self.balance = balance

Generate the complete code and Initialize BankAccount class with attributes like name, balance, Method to deposit amount, Method to withdraw amount, Method to check balance and display the output.



```
25
26      # Method to check balance
27      def check_balance(self):
28          print(f"Current balance: {self.balance}")
29
30  # Example usage
31  account1 = BankAccount("John Doe", 1000)
32  print(f"Account Holder: {account1.name}")
33  account1.check_balance()
34  account1.deposit(500)
35  account1.withdraw(200)
36  account1.withdraw(2000)
37  account1.check_balance()
38
```

```
PS C:\Users\kanch\OneDrive\Desktop\AI assisted coding>  c:; co
 coding'; & 'c:\Users\kanch\AppData\Local\Programs\Python\Pyth
ensions\ms-python.debugpy-2025.10.0-win32-x64\bundled\libs\deb
neDrive\Desktop\AI assisted coding\lab 4\lab 6.3\task5c.py'
Account Holder: John Doe
Current balance: 1000
500 deposited. New balance: 1500
200 withdrawn. New balance: 1300
Insufficient balance.
Current balance: 1300
PS C:\Users\kanch\OneDrive\Desktop\AI assisted coding>
```

Ctrl+K to generate a command

Search web & PC

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots

Evaluation Criteria:

Criteria	Max Marks
Class	1.0

	Loops	1.0		
	Conditional Statements	0.5		
	Total	**2.5 Marks**		